

# Energy Transfer in Collisions

## PART 1 • DESIGN YOUR OWN INVESTIGATION

Compact teacher guide • Reorganized from the original PLTW project for easier classroom use

**Part 1 Goal** Students act as scientists to investigate the testable question “What happens when objects collide?” They design a marble investigation, change one variable, conduct at least three trials, record observations, analyze evidence, and explain what they learned.

## At a Glance

|                                  |                                                                                            |
|----------------------------------|--------------------------------------------------------------------------------------------|
| <b>Testable question</b>         | What happens when objects collide?                                                         |
| <b>Core process</b>              | Ask → Predict → Investigate → Observe → Explain                                            |
| <b>Possible variables</b>        | Marble size • Track height • Release point                                                 |
| <b>Key vocabulary</b>            | scientist • scientific inquiry process • collision • variable • trend                      |
| <b>Evidence students produce</b> | Prediction • procedure • 3+ trials • sketches/observations • claim supported with evidence |

## Materials

- 2 track pieces and 1 track connector per group
- 2 small marbles and 2 large marbles per group
- Measuring tools: ruler, measuring tape, yardstick, and/or meterstick
- PLTW Launch Logs and pencils/colored pencils
- Optional: device with a slow-motion video analysis tool

## Before Class

- Review the vocabulary and the Scientific Inquiry Process video.
- Gather measuring tools and test the slow-motion tool if you plan to use it.
- Plan how students will record their findings in their Launch Logs.
- Optional: prepare the Vehicle Restraint System demonstration.

## Teaching Sequence

### 1. Connect to Prior Learning

- Review Activity 3 and William Kamkwamba’s windmill.
- Ask how energy was important to the windmill, where the energy came from, and how energy transfer/conversion helped provide electricity.
- Review collaboration: benefits of working with others, personal strengths, and ways to become a stronger collaborator.

**Optional Extension** Conduct the Vehicle Restraint System demonstration. Connect it to the need to understand

energy transfer in collisions before designing a restraint system.

## 2. Introduce Scientists & Scientific Inquiry

- Tell students: Today you get to act as scientists.
- Ask: What do scientists do?
- Clarify that scientists ask questions, make observations, and investigate ideas to gain knowledge and/or solve problems.
- Show the Scientific Inquiry Process video and discuss why scientists use this process.
- Have students prepare a Launch Log page titled “Energy Transfer in Collisions.”

**Discussion Prompts** How does the scientific inquiry process help scientists? • Why is asking questions important? • What makes a question testable? • How can we use this process to investigate collisions?

## 3. ASK — Build a Testable Question

- Have partners brainstorm questions about collisions.
- Share questions as a class and identify which are testable. Help students revise questions when needed.
- Present the investigation question: What happens when objects collide?
- Students record the question in their Launch Logs.

## 4. PREDICT 1 — Make an Initial Prediction

- Show the available materials.
- Students predict what will happen when the marbles collide.
- Remind students to use prior knowledge about energy, energy transfer, and energy conversion.
- Students share predictions in a small group, then discuss as a class.

**Ask** What do you think will happen when objects collide? Why? • How could we test this question? • Why is it important to investigate it?

## 5. INVESTIGATE 1 — Choose ONE Variable

Explain that a fair investigation changes only **one variable** while other factors stay the same.

**Quick Example** Paper towel test: change the brand of paper towel, but keep the amount of liquid the same. If two variables change, you cannot tell which variable caused the result.

| Variable to Test | What Changes                               | Keep the Same                                     |
|------------------|--------------------------------------------|---------------------------------------------------|
| Marble size      | small/small, small/large, large/large      | track height; release point                       |
| Track height     | same height, one end +6 in, one end +12 in | release point; marble size                        |
| Release point    | top, 4 in lower, 8 in lower                | marble size; track height; one marble's placement |

**Important** Students should not test multiple variables at the same time. Their investigation may differ from the

examples above, as long as it is testable.

## 6. INVESTIGATE 2 — Write the Procedure

- Groups develop clear, step-by-step directions for their investigation.
- Require at least three trials so students can look for trends in the data.
- Introduce trend: a pattern found in data.
- Students write the procedure for each trial in their Launch Logs.

## 7. PREDICT 2 — Revise the Prediction

After groups design their investigation, have them make their prediction more specific to the variable they chose.

- Marble size: “I think that when a small marble collides with a large marble, ...”
- Track height: “I think that when we raise the height of one end of the track, ...”
- Release point: “I think that when a rolling marble collides with a still marble, ...”

## 8. INVESTIGATE 3 — Set Up & Practice

- Groups gather materials and set up the investigation.
- Students practice releasing the marbles before Trial 1.
- Suggested roles: two students hold the ends of the track; two students release the marbles at the same time.
- Make sure marbles are lined up with each other for the planned collision.

## 9. OBSERVE — Record What Happens

- Students complete each trial and record what they see and hear.
- In the Launch Log, students create a table with a labeled sketch of the starting position and observations for each trial.
- Optional: record each collision with slow-motion video.
- Students analyze their data and use it as evidence to support a claim about collisions.

| Trial | Sketch of Starting Position | Observations |
|-------|-----------------------------|--------------|
| 1     |                             |              |
| 2     |                             |              |
| 3     |                             |              |

## 10. EXPLAIN — Share Findings & Use Evidence

- Pair each group with another group, ideally one that tested a different variable.
- Each group shares: the investigation they designed, their predictions, and their results.
- Lead a class discussion about what happens when objects collide.
- Ask students to support claims with evidence from their investigation.

**Evidence-Based Discussion** What happens when objects collide? • How did your variable affect the collision? • How is energy transferred in a collision—by sound, light, heat, force, or electric current? • What evidence supports your claim? • What new investigation would you design?

## Likely Evidence Students May Notice

- Objects' motion can change after a collision.
- A collision can produce sound.
- When a moving marble hits a still marble, the still marble can begin moving.
- Students may use changes in motion as evidence of energy transfer by force and the sound they hear as evidence of energy transfer by sound.

## Teacher Look-Fors

- The group changes only one variable.
- The procedure is clear enough for another group to follow.
- Students complete at least three trials.
- Observations describe what students actually see/hear rather than only what they expected.
- Students identify patterns/trends across trials.
- Claims are connected to specific evidence from the investigation.
- Students can explain that a collision can change objects' motion and transfer energy.

## Key Vocabulary

| Term                              | Student-Friendly Meaning                                                                                         |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>scientist</b>                  | A person who asks questions, makes observations, and investigates ideas to gain knowledge and/or solve problems. |
| <b>scientific inquiry process</b> | A process scientists use to explore observations and answer questions.                                           |
| <b>collision</b>                  | When objects hit one another.                                                                                    |
| <b>variable</b>                   | Something that can be controlled, measured, or changed in an investigation.                                      |
| <b>trend</b>                      | A pattern found in data.                                                                                         |

**Source Scope** This guide focuses only on Part 1: Design Your Own Investigation. Part 2: New Questions and the later Conclusion Questions are intentionally omitted.